🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

**UNIT 10 · STANDARD 6.GR.4**

# Surface Area Using Nets

Lesson 10-3 · Teacher Copy

**Name:** Type your name **Date:** Today's date **Class:** Class period

## My Notes Level 1 Support

🎯 **TODAY'S OBJECTIVES**

**Content Objective: I can find the surface area of a solid by using its net.**
**Language Objective: I can explain my work using the words surface area, net, face, and two-dimensional.**

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Surface area | Net | Face | Two-dimensional | Edge

👆 *Tap any word to see what it means and a picture.*

1. The total area of all the faces of a solid is its ______.
2. A flat pattern that folds up into a solid is a ______.
3. A flat surface of a solid figure is a ______.
4. A flat shape with length and width but no thickness is ______.
5. A line segment where two faces meet is an ______.

## Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

EXPLAIN

**For the box $l = 8$, $w = 5$, $h = 3$, you found three face pairs. How did you use the net to organize the faces into pairs, and which pair had the largest area?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Surface area** — The total area of all the flat sides of a solid.
  *Área de superficie — El área total de todos los lados planos de un sólido.*
- ☐ **Net** — A flat shape that folds up into a solid.
  *Plantilla (desarrollo plano) — Una figura plana que se dobla y forma un sólido.*
- ☐ **Face** — One flat side of a solid shape.
  *Cara — Un lado plano de una figura sólida.*
- ☐ **Two-dimensional** — A flat shape with length and width, but no thickness.
  *Bidimensional — Una figura plana con largo y ancho, pero sin grosor.*
- ☐ **Edge** — The line where two flat sides of a solid meet.
  *Arista — La línea donde se unen dos lados planos de un sólido.*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.
*Di tu idea a un compañero o en voz baja antes de escribir.*

**The faces come in ____ pairs because opposite faces are ____.**
*Las caras vienen en ____ pares porque las caras opuestas son ____.*

## 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** A net lays every face flat so you can see and measure all six faces.

**Evidence:** Students explain a rectangular prism has 6 faces, and the net unfolds the solid into a flat (two-dimensional) pattern so every face is visible to measure.

**Reasoning:** This evidence connects the definition of surface area to the mathematical claim.

### Start

**Most language support**

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **The faces come in ____ pairs because opposite faces are ____.**
  *Las caras vienen en ____ pares porque las caras opuestas son ____.*
- **The largest pair is ____ × ____ = ____, counted twice for ____ in².**
  *El par más grande es ____ × ____ = ____, contado dos veces para ____ in².*

### Build

**Some language support**

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is ____.**
  *Mi afirmación es ____.*
- **I know because ____.**
  *Lo sé porque ____.*
- **So ____.**
  *Entonces ____.*

### Explain

**Light language support**

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is ____.**
  *Mi afirmación es ____.*
- **My math evidence is ____.**
  *Mi evidencia matemática es ____.*
- **This proves ____ because ____.**
  *Esto demuestra ____ porque ____.*

## 4. Check Your Explanation

- [ ] I answered the question.
  *Respondí la pregunta.*
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  *Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*
- [ ] I explained how the evidence proves my answer.
  *Expliqué cómo la evidencia demuestra mi respuesta.*
- [ ] I used at least two math words.
  *Usé por lo menos dos palabras matemáticas.*
- [ ] I reread complete sentences and punctuation.
  *Volví a leer mis oraciones completas y la puntuación.*

## Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

## Answer Key & Teacher Guide

1. **Notes 1:** Surface area
2. **Notes 2:** Net
3. **Notes 3:** Face
4. **Notes 4:** Two-dimensional
5. **Notes 5:** Edge
6. **Watch for this mistake:** *A common mistake in Surface Area Using Nets is finding the area of only one face from each of the 3 matching pairs and forgetting to double it, instead of counting all 6 faces. For a box that is 9 in × 4 in × 5 in, a student might add just one face from each pair: (9×4) + (9×5) + (4×5) = 36 + 45 + 20 = 101 in², but the net actually has two of each face, so the correct total is 2(36) + 2(45) + 2(20) = 72 + 90 + 40 = 202 in² — exactly double. Before you submit, ask: 'Did I count both matching faces in each pair, or only one from each pair?'*
7. **Try It 1:** A. 6 *— A rectangular prism has 6 faces — top, bottom, front, back, left, and right — so its net shows 6 faces.*
8. **Try It 2:** A. 52 in² *— From the net: SA = 2(2×3) + 2(2×4) + 2(3×4) = 12 + 16 + 24 = 52 in².*